








# Blockchain Applications

## Lecture 5: Stablecoins II – Systemic Depegging Pressure and Portfolios



Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
Master Blockchain for Business (MBB), FABIZ
25 November 2026

# Today's Question and Route

- **Question**: Lecture 4 watched one pool. Can we measure pressure on the *whole* stablecoin system in one number, and is that number useful for forecasting?
- **Route**
  - comparing tokens with different anchors: reference adjustment
  - quantile regression and the Lasso; the Financial Risk Meter (FRM) idea
  - the De-pegging Pressure Index (DPI): construction, regimes, events
  - inside the tail links: who is linked to whom
  - measure or forecast: two different tasks
- Main source: Fan, Sha, Härdle & Pele (2026), *Reference-adjusted Quantile Lasso reveals regime-dependent stablecoin depegging pressure*, unpublished draft (2026), no public version; based on Fan (2026), MSc thesis, Humboldt-Universität zu Berlin
- Own replication on End-of-Day Historical Data (EODHD), 10 stablecoins, 1 January 2020 – 23 October 2026; gold and EUR/USD references: Monday–Friday closes, Friday's close carried over the weekend

## From Lecture 4: Three Different Objects

- The SVB weekend, March 2023, seen through three different numbers
- **Observed depeg** (Lecture 4): a price fact, in basis points (bp)
  - USDC daily close on 11 March 2023: $d = -285$ bp against the dollar
- **Contemporaneous pressure** (this lecture): the De-pegging Pressure Index (DPI), a model-implied penalty with no unit
  - DPI on 12 March 2023, from the 90 days to 11 March: 0.52 (fourth of five quintiles, 0.43–0.55)
- **Future probability** (Lecture 4): an alarm's estimate of $P(\text{breach within the next 24 hours})$, a number in $[0, 1]$
- The three answer different questions: what happened, how strongly coins move together in their lower tails now, what may happen next
- The DPI is not a depeg probability; whether it helps to forecast is tested at the end

## Learning Outcomes

- Compute a reference-adjusted peg deviation for dollar, euro and gold tokens
- Write down the check loss, the 5% Quantile Lasso and the generalised approximate cross-validation (GACV) criterion, and explain what the selected penalty measures
- Construct and interpret the DPI, its regimes and event responses, and judge their robustness
- Distinguish in-sample fit from out-of-sample forecasting
- Optional (appendix): judge whether the DPI is useful for building a portfolio

# Abbreviations

**bp** basis point, 0.01%
**BUSD** Binance USD (stablecoin)
**CVIX** crypto volatility index
**DAI** Dai (MakerDAO stablecoin)
**DD** drawdown (max DD: maximum drawdown)
**df** degrees of freedom
**DPI** De-pegging Pressure Index
**DXY** US dollar index
**EODHD** End-of-Day Historical Data (data provider)
**ES** expected shortfall
**EUR** euro
**EURS** STASIS Euro (stablecoin)
**FABIZ** Faculty of Business Administration in Foreign Languages, ASE
**FRM** Financial Risk Meter
**FTX** FTX crypto exchange (collapsed November 2022)
**GACV** generalised approximate cross-validation
**GMV** global minimum variance (portfolio)
**GUSD** Gemini Dollar (stablecoin)
**HAC** heteroskedasticity and autocorrelation consistent
**HHI** Herfindahl–Hirschman index
**IDRT** Rupiah Token (stablecoin)
**L1** L1 norm, $\|\beta\|_1 = \sum_k |\beta_k|$ (Lasso penalty)
**Lasso** least absolute shrinkage and selection operator
**MAD** median absolute deviation
**n.s.** not significant
**OOS** out-of-sample
**PAXG** Pax Gold (gold-backed token)
**pp** percentage points
**S&P 500** Standard & Poor's 500 stock index
**SD** standard deviation
**SE** standard error
**sUSD** Synthetix USD (stablecoin)
**SVB** Silicon Valley Bank (failed March 2023)
**TUSD** TrueUSD (stablecoin)
**USD** US dollar
**USDC** USD Coin (stablecoin)
**USDP** Pax Dollar (stablecoin)
**USDT** Tether (stablecoin)
**VIX** CBOE volatility index
**XAU** gold, one troy ounce

## Reading and Tools

- Fan, Sha, Härdle & Pele (2026), unpublished draft; Fan (2026), MSc thesis, Humboldt-Universität zu Berlin, *Large Quantile Lasso FRM for Stable Coin Depegging Risks*, not public; keynote *Stable Coin Risks* (Fan, Härdle, Pele, Sha, 16 September 2026); results marked "paper" are quoted from the draft
- FRM: Mihoci, Althof, Chen & Härdle (2020), Yu, Härdle, Borke & Benschop (2023); tail networks: Härdle, Wang & Yu (2016), Adrian & Brunnermeier (2016)
- Methods: Koenker & Bassett (1978) (quantile regression), Tibshirani (1996) (Lasso), Belloni & Chernozhukov (2011) (L1 quantile regression)
- Stablecoin arbitrage: Lyons & Viswanath-Natraj (2023)
- Video unit on Quantinar: FRM for Cryptos
- Estimation: Python (`scipy` linear programming, `joblib`); more than 48 000 Quantile Lasso equations, each on a grid of 25 penalties, estimated for this lecture

## Reference Adjustment

- Not every stablecoin targets the dollar: EURS tracks the euro, PAXG one ounce of gold, IDRT the rupiah
- Let $P_{i,t}$ be the dollar price of token $i$ and $B_{i,t}$ the dollar value of its reference asset:

$$d_{i,t} = 10^4 \left( \frac{P_{i,t}}{B_{i,t}} - 1 \right) \text{ bp}, \qquad r_{i,t} = 10^4 \, \Delta \ln \frac{P_{i,t}}{B_{i,t}}$$

  - $B = 1$ for dollar tokens; EUR/USD for EURS; XAU/USD for PAXG
- Worked example: EURS at 1.1194 USD while EUR/USD = 1.1202

$$d = 10^4 \left( \frac{1.1194}{1.1202} - 1 \right) = -7.1 \text{ bp}$$

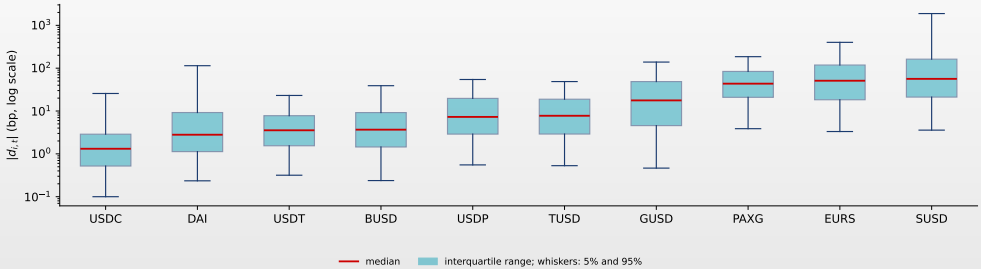
  - without adjustment, a 1% move of the euro would look like a 100 bp "depeg" of EURS
- The adjustment puts all tokens on one economic scale *before* any model sees them

# Peg Quality, January 2020 – May 2026

| coin | med $\lvert d\rvert$ | own (EODHD) $q_{0.05}$ | own (EODHD) % $d \le -100$ | paper med $\lvert d\rvert$ | coin | med $\lvert d\rvert$ | own (EODHD) $q_{0.05}$ | own (EODHD) % $d \le -100$ |
|---|---|---|---|---|---|---|---|---|
| USDC | 1.3 | −5.0 | 0.17 | 4.11 | GUSD | 17.7 | −110.0 | 6.32 |
| DAI | 2.8 | −9.0 | 0.68 | 7.06 | PAXG | 43.5 | −124.4 | 9.13 |
| USDT | 3.6 | −11.3 | 0.09 | 5.37 | EURS | 51.0 | −251.9 | 17.11 |
| BUSD | 3.7 | −8.6 | 0.09 | | sUSD | 56.3 | −1 883.4 | 25.99 |
| USDP | 7.3 | −47.5 | 0.21 | 7.63 | | | | |
| TUSD | 7.8 | −37.8 | 2.09 | | | | | |

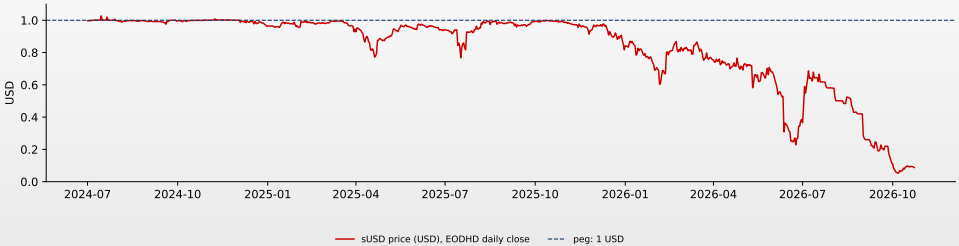
- $d$ in bp; % $d \le -100$: share of days at least 100 bp below the reference; 2 343 days per coin, EURS 2 315 (a 43-day gap in the EODHD file, November–December 2025)
- Two tiers: large dollar tokens within a few bp; small, non-dollar or crypto-collateralised tokens tens of bp off, with deep lower tails
- The source matters for levels (EODHD medians are lower than CoinGecko's), not for the ranking; the downside rates agree closely (USDT 0.09% in both; sUSD 26.0% vs 25.0%)
- sUSD (crypto-collateralised): 0.95 USD in March 2025, 0.76 in March 2026, 0.09 on 23 October 2026, a persistent depeg

## Two Tiers of Peg Quality (own)




- Distribution of $|d_{i,t}|$, January 2020 – May 2026, coins ordered by median; log scale
- The 95% whisker spans three orders of magnitude: 25 bp for USDC, almost 2 000 bp for sUSD

# A Persistent Depeg: sUSD (own)



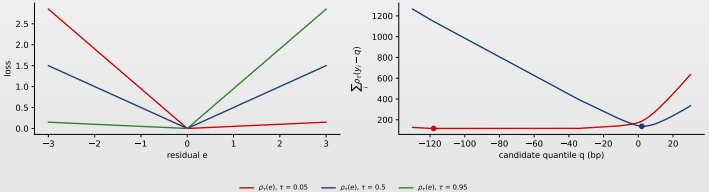

- Crypto-collateralised (Synthetix); below 0.99 USD on 607 of 661 days since January 2025; below 0.95 from 24 November 2025; 0.09 USD on 23 October 2026
- Unlike Silicon Valley Bank (SVB; two days), no arbitrage closes this gap: redemption at par is not available; a depeg can be a slow default

## The Check Loss

- The $\tau$-quantile minimises an asymmetric absolute loss (Koenker & Bassett (1978)):

$$\rho_\tau(e) = e\,\big(\tau - \mathbb{1}\{e < 0\}\big), \qquad q_\tau = \arg\min_q E\,\rho_\tau(Y - q)$$

- Quantile regression: $\min_{\alpha,\beta} \sum_i \rho_\tau(y_i - \alpha - x_i'\beta)$, a linear programme
- For $\tau = 0.05$ a residual below the line costs $0.95|e|$, above it $0.05|e|$: the fit is pushed to the lower tail




- Right: 20 daily deviations; for $\tau = 0.5$ the minimum is the median (2 bp); for $\tau = 0.05$ the loss is *flat* between $-120$ and $-35$ bp

## Worked Example: How Many Points Define a 5% Quantile?

- With $n = 20$ and $\tau = 0.05$: $n\tau = 1$, so one observation lies below the quantile

$$\sum_i \rho_{0.05}(y_i - q) = 116.55 \quad \text{for every } q \in [-120, -35]$$

- Any value between the smallest and second-smallest observation is a valid 5% quantile: the estimate is not unique
- In a 90-day window $n\tau = 4.5$ is not an integer: the minimiser is unique, the 5th smallest observation; still, five days carry the whole lower tail
- Interpretation
  - rolling 5% quantile models are noisy by construction; one day entering or leaving the window can move the estimate
  - this is why the DPI is analysed in regimes and event windows, not day by day

## Quantile Lasso and the Selected Penalty

- With many candidate predictors, add an L1 penalty (Tibshirani (1996), Belloni & Chernozhukov (2011)):

$$(\hat{\alpha}, \hat{\beta})(\lambda) = \arg\min_{\alpha,\beta} \sum_{s \in W_t} \rho_{0.05}\left(d_{j,s} - \alpha - x'_{j,s}\beta\right) + \lambda \|\beta\|_1$$

- All slopes are zero for $\lambda \geq \lambda_{max}$, with $(\psi_\tau(u) = \tau - \mathbb{1}\{u < 0\})$

$$\lambda_{max} = \max_k \left| \sum_s x_{k,s}\, \psi_\tau\left(d_{j,s} - \hat{q}_\tau\right) \right|$$

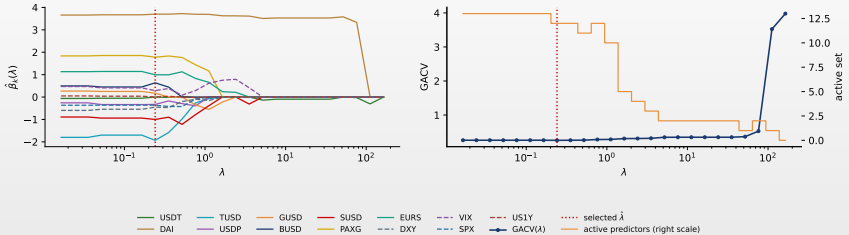
  - a scaled covariance between each predictor and the target's lower-tail indicator

- $\lambda$ is chosen by generalised approximate cross-validation on a grid:

$$\mathrm{GACV}(\lambda) = \frac{\sum_s \rho_\tau\left(d_{j,s} - \hat{\alpha} - x'_{j,s}\hat{\beta}(\lambda)\right)}{n - \mathrm{df}(\lambda)}, \qquad \mathrm{df} = \#\{\hat{\beta}_k \neq 0\} + 1$$

- FRM idea (Mihoci, Althof, Chen & Härdle (2020)): the selected $\hat{\lambda}_{j,t}$ is used as a risk proxy
  - its level also depends on scaling, noise, the $\lambda$ grid and the selection criterion: a specification-dependent proxy, to be checked against other stress measures

# Worked Example: One Equation, USDC after SVB (own)




- Target USDC, 90 days to 19 March 2023, 13 candidates (9 coins, 4 lagged macro changes), all scaled by the median absolute deviation (MAD)
- $\lambda_{max} = 164.1$; GACV minimum at $\hat{\lambda} = 0.241$ with 12 active predictors; largest slopes: DAI 3.70, TUSD $-1.93$, PAXG 1.78
- DAI enters first and stays to the largest $\lambda$: in the SVB window USDC's lower tail was DAI's lower tail

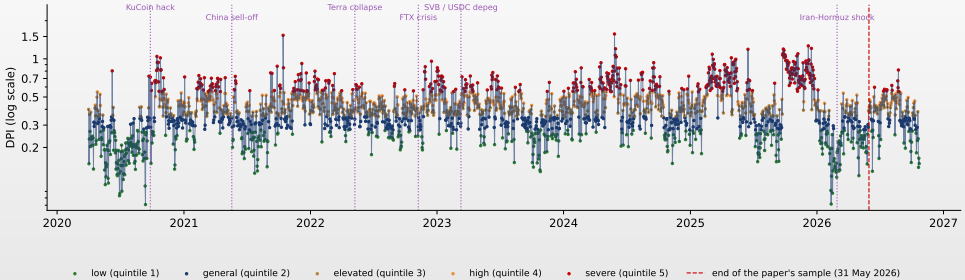
## Construction

- For each coin $j$ and day $t$, a 5% Quantile Lasso on the preceding 90 days
  - target: $d_{j,s}$; predictors: the other coins' $d_{k,s}$ (same day) and one-day-lagged changes of the US dollar index (DXY), the CBOE volatility index (VIX), S&P 500, 1-year Treasury yield (paper also: CVIX)
  - every variable centred by its window median and divided by $1.4826 \times \mathrm{MAD}$
  - 25 values of $\lambda$; $\hat{\lambda}_{j,t}$ minimises GACV
  - real time: the window uses only prices known on day $t$ (the last day as recorded, before any data check)
- Index:

$$\mathrm{DPI}_t = \frac{1}{J} \sum_{j=1}^{J} \hat{\lambda}_{j,t}$$

- Paper: 11 coins (CoinGecko), 2 252 days to 31 May 2026. Own: 10 coins (IDRT has no EODHD data after 2023), 2 397 days to 23 October 2026, 23 970 equations
- The DPI is *not* a depeg probability or a percentage; it is a model-implied measure of lower-tail dependence

# The DPI, April 2020 – October 2026 (own)




- Annual medians: 0.27 (2020), 0.39–0.40 (2021–2023), 0.44 (2024), 0.52 (2025), 0.32 (2026); highest regime in late 2025, at the start of the sUSD depeg
- Log-linear trend to May 2026: $+0.046$ per year, heteroskedasticity and autocorrelation consistent (HAC(90)) standard error (SE) 0.036, $p = 0.21$, 95% CI $[-0.025, 0.117]$
- We do not reject a zero slope in this specification; the interval is wide, so this is not evidence of no trend
- Paper: $-0.004$ per year, $p = 0.78$: the two agree only in not rejecting zero; the signs differ

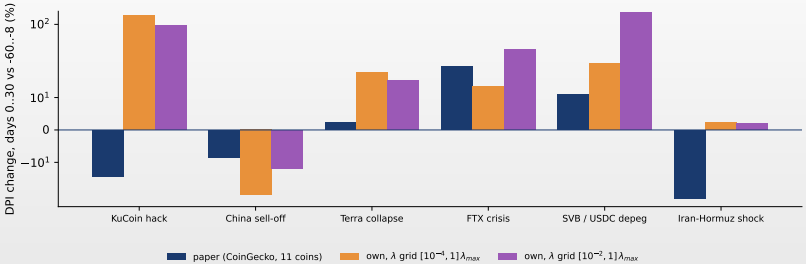
## Event Responses

- Response: $\Delta = \overline{\mathrm{DPI}}_{[0,30]} / \overline{\mathrm{DPI}}_{[-60,-8]} - 1$; test on $\ln \mathrm{DPI}$ with HAC errors; Holm correction for six events

| Event | date | paper | own | own Holm p | own, narrow $\lambda$ grid |
|---|---|---|---|---|---|
| KuCoin hack | 26 Sep 2020 | $-14.4\%$ | $+141.1\%$ | $< 0.001$ | $+95.3\%$ |
| China sell-off | 19 May 2021 | $-8.6\%$ | $-20.2\%$ | 0.033 | $-11.9\%$ |
| Terra collapse | 9 May 2022 | $+2.2\%$ (n.s.) | $+17.9\%$ | 0.021 | $+15.4\%$ |
| FTX crisis | 8 Nov 2022 | $+19.5\%$ | $+13.4\%$ | 0.961 | $+36.7\%$ |
| SVB / USDC depeg | 11 Mar 2023 | $+11.0\%$ | $+21.0\%$ | 0.092 | $+158.6\%$ |
| Iran–Hormuz shock | 28 Feb 2026 | $-22.9\%$ | $+2.4\%$ | 0.961 | $+1.9\%$ |

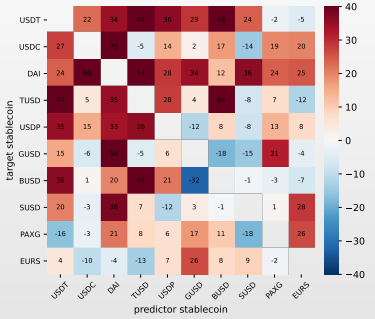
- Paper: significant rises only after FTX and SVB, the two events about intermediation and reserve confidence
- Own replication: the sign agrees for 4 of 6 events, but sizes and significance change with the data source and even with the $\lambda$ grid

## How Robust Are Event Responses?




- Narrow grid $[10^{-\mathbf{2}}, 1]\lambda_{max}$: 19.6% of equations select the smallest $\lambda$ (a boundary solution); wide grid $[10^{-\mathbf{4}}, 1]\lambda_{max}$: 4.0%
- The two indices correlate 0.84 in logs, yet the SVB response is $+21\%$ or $+159\%$: a 31-day mean of a noisy index is sensitive to tuning choices
- Lesson: report event responses with their specification; trust the results that survive several specifications

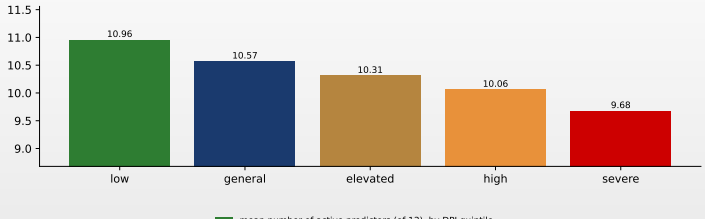
# Who Is Linked to Whom? (own)




signed selection frequency (pp): % of daily equations with a positive slope minus % with a negative slope; not a transmission size

- Positive slope = *aligned*: a lower predictor deviation goes with a lower target quantile
- Cell = signed selection frequency: % of daily equations with a positive slope minus % with a negative slope, in percentage points (pp); not a size in bp
- Aligned 47.4% vs inverse 33.1% of coin coefficients (paper 39.6% vs 23.9%)
- Most often aligned: USDC ← DAI (75 pp), TUSD ← BUSD (64), DAI ← USDC (60); most often inverse: BUSD ← GUSD (−32)
- Macro variables: 18.4% of absolute slope weight (paper 20.8%)
- Paper: no pair difference survives multiple testing; rankings change between halves: common structure, no fixed sender

# Sparser Models under Pressure




- Own: from 10.96 active predictors (of 13) in the lowest DPI quintile to 9.68 in the highest; paper: 10.11 to 9.00 (of 15)
- Partly mechanical: a larger selected penalty removes predictors by construction
- Economic reading (a few strong tail links dominate under pressure): a hypothesis, not shown by this chart
- Same direction in both data sets: one of the results that survives replication
- Check against simple stress measures (90-day means): Spearman correlation of the DPI with the median $|d|$ across coins $-0.10$, with the VIX $-0.16$: the DPI is not a proxy for either

# Fit Is Not Forecast (Paper)

| Model | in-window GACV skill | same-day OOS nowcast skill | fully lagged OOS forecast skill |
|---|---|---|---|
| macro only | 8.3% | −12.5% | −12.5% |
| coins only | 33.7% | −15.9% | −34.4% |
| joint | 37.0% | −19.8% | −38.6% |

- Skill = 1− loss of the model / loss of the intercept-only model, on the paper's sample (to 31 May 2026)
  - in-window: GACV inside each 90-day estimation window, not a held-out loss
  - OOS: check loss on days outside the estimation window; nowcast: coin predictors from the target's own day; forecast: every predictor dated $t-1$; benchmark: rolling intercept-only model
- The best in-window description (joint, 37%) is the worst forecast once predictors are dated $t-1$ (−38.6%)
- The DPI alone has negative out-of-sample (OOS) $R^2$ at horizons of 10 to 150 days; the crypto FRM forecasts crypto volatility ($R^2$ 0.17–0.29) but adding the DPI does not help
- On-chain variables (Curve, Uniswap) enter 91% of equations, yet do not reduce held-out loss significantly: frequent selection is not forecast value
- The DPI and crypto FRM correlate only 0.11: stablecoin pressure is not crypto stress

# Concentration: Exposure, Not a Daily Alarm (Paper)

- Normalised Herfindahl–Hirschman index (HHI) of the paper's $J = 11$ coins: $(\mathrm{HHI} - 1/J)/(1 - 1/J)$, $\mathrm{HHI} = \sum_j s_j^2$, $s_j$ = market-value share within the 11 coins
  - 0.640 (2020), 0.291 (2022), 0.513 (2026); on 31 May 2026 USDT 69.4%, USDC 28.0% of the 11 coins
  - not comparable with the raw HHI of Lecture 1 or with shares of all USD stablecoins (Lecture 4): other universe and date
- Regression of $\ln \mathrm{DPI}_t$ on lagged concentration, controls and year effects (HAC(90)):
  - $+1$ standard deviation (SD) of HHI: $-1.52\%$ DPI $[-4.24\%, 1.27\%]$; Top-2 share: $-2.01\%$ $[-4.93\%, 1.01\%]$; Holm p $= 0.38$
- Without year effects, higher concentration goes with higher DPI; with them the link disappears: both trend with the market regime
- Interpretation
  - concentration tells you *how much* of the system depends on two issuers: use it for scenarios and limits (Lecture 1: HHI)
  - it does not tell you *when* pressure builds

## Measure or Forecast?

| Task | Evidence | Verdict |
|---|---|---|
| Measure the current tail structure | aligned $>$ inverse links; regimes, zero trend not rejected | DPI useful, as one layer of monitoring |
| Forecast | negative OOS skill; lagged models worse; DPI adds nothing to FRM | no |
| Event attribution | FTX and SVB in the paper; sizes change with source and $\lambda$ grid | fragile: report the specification |

- A state measure is not a leading indicator: test each claim separately

## Summary

- Reference adjustment puts dollar, euro and gold tokens on one scale; peg quality splits into two tiers (USDC 1.3 bp median vs EURS 51 bp, sUSD 56 bp)
- A 5% quantile in a 90-day window is the 5th lowest observation: rolling tail models are noisy by construction
- The DPI averages GACV-selected Quantile Lasso penalties; it moves in regimes, and a zero trend is not rejected ($+0.046$ a year, 95% CI $[-0.025, 0.117]$)
- Good fit is not forecast: held-out skill is negative, and worse with more information

## Interpretation Exercises

1. PAXG trades at 3 998 USD while gold is 3 994 USD. Compute nothing: explain why its dollar return is not a depeg measure and what the reference adjustment changes.
2. In a 20-day window the 5% check loss is flat between the two lowest observations. What does this imply for a rolling 5% quantile, and what changes in a 90-day window?
3. The DPI rose by 159% after SVB in one specification and by 21% in another. Which results of the lecture would you still report with confidence, and why?
4. The HHI coefficient is $-1.52\%$ with interval $[-4.24\%, 1.27\%]$ after year effects. What is the conclusion, and what is concentration still useful for?

- Next: Lecture 6, networks, sentiment and asset classes; there a "link" becomes a correlation or a variance share, not a selection frequency

# Appendix: Further Material

Optional material for self-study. It is **not examined**.

- Data Check: Nine Bad Prints (own)
- Coins or Macro? A Shapley Decomposition
- Four Long-Only Rules
- Results: Covariance Beats Penalties
- Value and Drawdown, 2020–2026 (own)
- What the Minimum-Variance Portfolio Holds (own)
- Worked Example: Why the Penalty Is a Bad Weight

# Data Check: Nine Bad Prints (own)

- Rule (as in Lecture 1): a daily move beyond 1 000 bp that reverses by at least 90% the next day is treated as a recording error; the close is set to missing and filled from the past
- The rule needs the *next* day: a retrospective cleaning, used for the descriptive tables; the DPI and the portfolio weights on day $t$ see day $t-1$ as recorded

| coin | date | close (USD) | log return (bp) |
|---|---|---|---|
| sUSD | 6 Aug 2025 | 0.2743 | −12 446 |
| EURS | 15 Jun 2025 | 0.8788 | −2 720 |
| sUSD | 6 Oct 2026 | 0.0724 | +2 633 |
| EURS | 23 Jun 2025 | 0.9362 | −1 953 |
| sUSD | 14 Feb 2020 | 0.8372 | −1 566 |
| four more | 2020–2026 | | 1 000 to 1 500 |

- sUSD on 6 August 2025: low 0.21, close 0.27, next open 0.27, next close 0.98, on low volume; a reversal alone does not prove an error, and no second source is available here
- Sensitivity of the DPI: retrospective instead of real-time cleaning changes 6 days, the trend and event responses not at all; no cleaning changes 437 days, trend +0.041 ($p = 0.24$), KuCoin +153% instead of +141%
- Sensitivity of the portfolios (2020 – May 2026), returns from uncleaned prices: linear-lambda maximum drawdown 29.2% instead of 19.0%, equal-weight volatility 6.2% instead of 3.9%

## Coins or Macro? A Shapley Decomposition

- Conditional skill of a model $m$: $S_m = 1 - \overline{\mathrm{GACV}}_m / \overline{\mathrm{GACV}}_0$ (0 = intercept only)
- Two-block Shapley value of the coin block:

$$\phi_{coin} = \tfrac{1}{2}\left[(S_{coin} - S_0) + (S_{joint} - S_{macro})\right], \qquad \phi_{coin} + \phi_{macro} = S_{joint}$$

| | paper | | own (every 7th day) | |
|---|---|---|---|---|
| Model | GACV | skill | GACV | skill |
| intercept only | 0.1808 | – | 0.1679 | – |
| macro only | 0.1658 | 8.3% | 0.1577 | 6.1% |
| coins only | 0.1199 | 33.7% | 0.1033 | 38.5% |
| joint | 0.1139 | 37.0% | 0.0980 | 41.6% |
| coin share of joint skill $\phi_{coin}/S_{joint}$ | | 84.3% | | 88.9% |

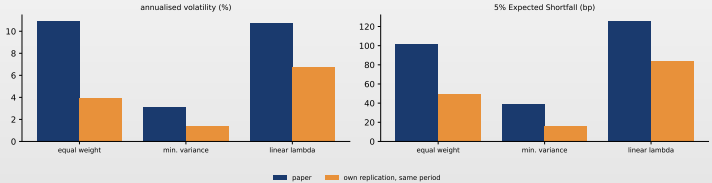
- Worked check (own): $\phi_{coin} = \tfrac{1}{2}[(38.5 - 0) + (41.6 - 6.1)] = 37.0$; $37.0/41.6 \approx 0.89$
- In-window (GACV) result, robust across the two data sets: the other stablecoins explain most of a coin's lower tail; macro variables add little

## Four Long-Only Rules

- Holding reference-adjusted stablecoin positions (a treasury, a market maker, a lending desk); returns $r_{i,t}$ in bp
- Rules, rebalanced every 7 days with information to $t-1$, weights capped at 30%:
  - equal weight: $w_i = 1/J$
  - global minimum variance (GMV): $\min_w w'\hat{\Sigma}w$, $\sum w_i = 1$, $0 \le w_i \le 0.3$; $\hat{\Sigma} = 0.9\,S + 0.1\,\mathrm{diag}(S)$, $S$ the covariance of the last 90 days (10% shrinkage)
  - linear lambda: $\min_w \sum_i w_i \hat{\lambda}_{i,t-1}$ under the same constraints (fill the lowest-penalty coins first)
  - quadratic lambda (paper only): penalty level plus covariance
- Holdings simulation: weights drift with daily simple returns $e^{r_{i,t}/10^4} - 1$ between rebalances; turnover $= \frac{1}{2}\sum_i |w_i^{target} - w_i^{drifted}|$; no transaction costs
- Outcomes: annualised mean return and volatility, 5% ES of daily returns, maximum drawdown, turnover per rebalance

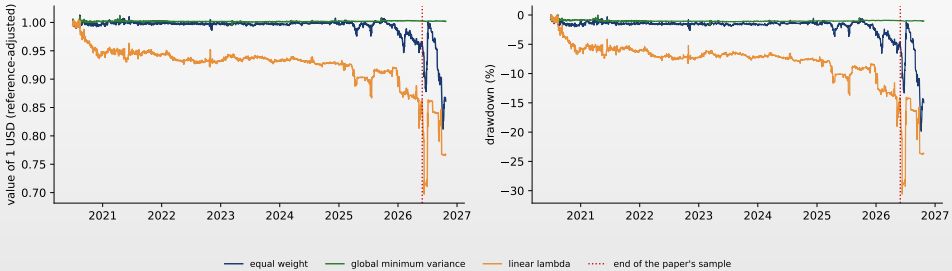
# Results: Covariance Beats Penalties

| | own, 30 Jun 2020 – 31 May 2026 | | | | | paper | | |
|---|---|---|---|---|---|---|---|---|
| Rule | return | vol | ES 5% (bp) | max DD | turnover | vol | ES 5% | max DD |
| equal weight | −0.51% | 3.92% | 49.0 | 6.6% | 0.3% | 10.93% | 101.4 | 15.6% |
| min. variance | 0.05% | 1.43% | 15.9 | 1.7% | 3.7% | 3.14% | 39.0 | 2.6% |
| linear lambda | −2.46% | 6.72% | 84.2 | 19.0% | 57.0% | 10.76% | 125.7 | 16.5% |



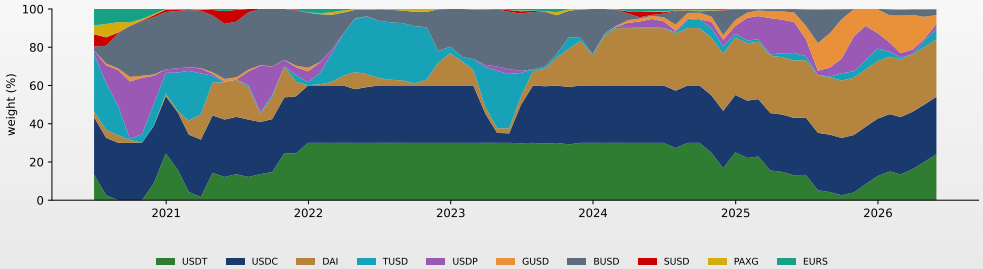

- Robust result: GMV cuts volatility by 63% and ES by 67% against equal weight (paper: 71%, 62%); penalty-based weights add risk and turnover

# Value and Drawdown, 2020–2026 (own)




- GMV stays within 1.7% of its peak over six years; equal weight falls 19.9% below its peak once sUSD collapses (after the paper's sample)
- Linear lambda loses about 6% in its first months (2020) and more in each stress episode: risk without a reward

# What the Minimum-Variance Portfolio Holds (own)




- Average weights: USDC 28.8%, USDT 21.1%, DAI 16.9%, BUSD 14.9%, TUSD 8.2%; sUSD, PAXG, EURS below 1% each
- The cap binds often: without it GMV would hold even more USDC and USDT, i.e. more issuer concentration

# Worked Example: Why the Penalty Is a Bad Weight

- $\hat{\lambda}_{j,t}$ is the regularisation chosen for one equation; it depends on how the *other* coins move in $j$'s lower tail, not on $j$'s own risk
- A small, erratic token (high variance, few co-movements) can get a *low* $\hat{\lambda}$ and then a 30% weight under the linear rule
- In numbers (own): linear lambda volatility 6.7% vs 1.4% for GMV; turnover 57.0% per week: the ranking of $\hat{\lambda}$ changes almost every rebalance
- Extension to 23 October 2026 (sUSD falls from 0.39 to 0.09):

| 30 Jun 2020 – 23 Oct 2026 | vol | ES 5% (bp) | max DD |
|---|---|---|---|
| equal weight | 5.74% | 70.5 | 19.9% |
| min. variance | 1.39% | 15.3 | 1.7% |
| linear lambda | 10.62% | 111.4 | 30.6% |

- Equal weight triples its drawdown (6.6% to 19.9%) with sUSD's collapse; GMV had already moved out because sUSD's 90-day volatility exploded (364 bp a day before 12 June 2026)